NAME:T.JASHWANTH

REG;NO:192572109

9. Scenario: You work for a real estate agency and have been given a dataset containing information about properties for sale. The dataset is stored in a Pandas DataFrame named property_data. The DataFrame has columns for property ID, location, number of bedrooms, area in square feet, and listing price. Your task is to analyze the data and answer specific questions about the properties.

Question: Using Pandas DataFrame operations, how would you find the following information from the property_data DataFrame:

1.The average listing price of properties in each location.

2.The number of properties with more than four bedrooms.

3.The property with the largest area.

CODE:

```
import pandas as pd

data = {
    'Property_ID': [101, 102, 103, 104],
    'Location': ['Chennai', 'Hyderabad', 'Chennai', 'Bangalore'],
    'Bedrooms': [3, 5, 2, 6],
    'Area': [1500, 2500, 1200, 3000],
    'Price': [5000000, 8000000, 4000000, 9000000]
}

property_data = pd.DataFrame(data)

# 1. Average price in each location
```

```
print(property_data.groupby('Location')['Price'].mean())

# 2. Properties with more than 4 bedrooms
print(property_data[property_data['Bedrooms'] > 4])

# 3. Property with the largest area
print(property_data.loc[property_data['Area'].idxmax()])
```

INPUT:

Property_ID: [101, 102, 103, 104]

Location: ['Chennai', 'Hyderabad', 'Chennai', 'Bangalore']

Bedrooms: [3, 5, 2, 6]

Area: [1500, 2500, 1200, 3000]

Price: [5000000, 8000000, 4000000, 9000000]

OUTPUT:

```
Location
Bangalore    9000000.0
Chennai      4500000.0
Hyderabad    8000000.0
Name: Price, dtype: float64
   Property_ID   Location  Bedrooms  Area    Price
1          102  Hyderabad         5  2500  8000000
3          104  Bangalore         6  3000  9000000
Property_ID          104
Location       Bangalore
Bedrooms               6
Area                3000
Price            9000000
Name: 3, dtype: object
```